theory as what you get when you quantize a classical field theory in a preexisting space.

One of the lessons we will learn is that entanglement plays an even more central role in quantum field theory than it does in quantum particle theories. When particles were our primary concern, entanglement was something that may or may not have been important, depending on the physical circumstances. You can create a state of two entangled electrons, but there are plenty of interesting states of two electrons where the particles aren't entangled at all. In field theory, by contrast, essentially every physically interesting state is one that features an enormous amount of entanglement. Even empty space, which you might think of as pretty straightforward, is described in quantum field theory as an intricate collection of entangled vibrations.

Quantum mechanics first began when Planck and Einstein argued that electromagnetic waves had particle-like properties, and then Bohr, de Broglie, and Schrödinger suggested that particles could have wave-like aspects. But there are two different kinds of "waviness" at work here, and it's worth being careful to distinguish between them. One kind of waviness arises when we make the transition from a classical theory of particles to a quantum version, obtaining the quantum wave function of a set of particles. The other kind is when we have a classical field theory to start with, even before quantum mechanics becomes involved at all. That's the case with classical electromagnetism, or with Einstein's theory of gravity. Classical electromagnetism and general relativity are both theories of fields (and therefore of waves), but are themselves perfectly classical.

In quantum field theory, we start with a classical theory of fields and construct a quantum version of *that*. Instead of a wave function that tells us the probability of seeing a particle at some location, we have a wave function that tells us the probability of seeing a particular configuration of a field throughout space. A wave function of a wave, if you like.

There are many ways to quantize a classical theory, but the most direct one is the route we have already taken. Thinking of a collection of particles, we can ask, "Where can the particles be?" The answer for each individual